

GROUP 18 CASE STUDY ASSIGNMENT

The Fresh Valley Refrigeration Deal

Iqhwa Cold Chain Systems (Pty) Ltd and the Lead-to-Cash Process — Sales and Service

A group project based on two units of the SAP Learning course "Exploring End-to-End Business Processes in SAP Business Suite"

Assigned to	Group 18
Assessment weight	100 marks
Assessment type	Group written report, with process diagrams and appendices
Based on	SAP Learning, "Exploring End-to-End Business Processes in SAP Business Suite" — the units Performing the Lead-to-Cash Process: Focus on Sales, and Performing the Lead-to-Cash Process: Focus on Service (learning.sap.com)
Recommended length	3,000–4,000 words, excluding diagrams, tables and appendices
Submission format	Single PDF file to makombef@dut.ac.za by the Group Leader
Due date	5 October 2026

Assignment at a Glance

Iqhwa Cold Chain Systems (Pty) Ltd sold 75 Iqhwa WI-500 units to Fresh Valley Supermarkets (Appendix A) and signed a 3-year contract to maintain them (Appendix B). Your group's job is to map both halves of this one deal onto SAP, using the process names, documents and terms taught in the course.

Your group submits ONE report containing six parts. The table below is a checklist — full instructions for each part are in Section 5.

Part	You Submit	Marks
A	A table mapping all 10 Sales events, plus 3 written justifications	20
B	1 sales process diagram + 2 tables (master data, enterprise structure)	20
C	A table mapping all 11 Service events, plus 3 written justifications	20
D	2 service process diagrams + 1 table + 1 short comparison paragraph	20
E	1 integration explanation + 2 risk scenarios + 1 reflection (≤ 300 words)	15
F	Applies to the whole report — no separate item to write	5
Total		100
Before you start		
Read Appendix A and Appendix B (the two event logs, just after Section 3) first. Every task in Section 5 refers back to specific event numbers, so your group needs both logs in front of you before you begin.		

1. Introduction and Purpose

This assignment asks your group to act as a short-term consulting team engaged by Iqhwa Cold Chain Systems (Pty) Ltd, a commercial refrigeration manufacturer, to analyse a single major customer deal from first contact through to ongoing after-sales support — and to show, step by step, how that deal would be represented in SAP. Unlike a broad survey of every business function, this assignment focuses deliberately on two connected units of the SAP Learning course Exploring End-to-End Business Processes in SAP Business Suite: Performing the Lead-to-Cash Process: Focus on Sales, and Performing the Lead-to-Cash Process: Focus on Service.

These two units are a natural pair. The Sales unit covers how a company turns a lead into revenue — presales, order management, delivery, billing and payment. The Service unit covers what happens next: how a company supports what it has already sold, through service contracts, scheduled maintenance and on-demand repair, and bills for that ongoing relationship. Together they tell the complete revenue story of a product that is not just sold once, but sold and then supported for years. Your task is to trace both halves of that story for one deal, and to explain — concretely, in SAP terms — how the "sales" side and the "service" side of the same customer relationship connect.

1.1 Learning Outcomes

On successful completion of this assignment, your group will be able to:

- Explain the full sequence of the Lead-to-Cash Sales process, from presales through to payment.
- Explain the full sequence of the Lead-to-Cash Service process, distinguishing planned/recurring maintenance from unplanned/corrective maintenance.
- Map narrative business events to the correct SAP process stage, document, and responsible role or system.
- Identify the master data, enterprise structure and organizational structure elements each process depends on.
- Construct process flow diagrams that show the sequence of steps and the SAP objects each step creates or consumes.
- Explain how the Sales and Service processes integrate around a single customer and product, and evaluate the consequences of running them on disconnected systems.
- Collaborate as a group to produce a coherent, professionally presented business analysis report.

2. Required Background Reading

Before your group begins drafting, every member should complete (or revise) the two units of the SAP Learning course Exploring End-to-End Business Processes in SAP Business Suite listed below. The course is freely available at learning.sap.com. You are not required to complete any hands-on exercises, but you must understand the concepts, terminology and process steps taught in each lesson.

Unit: Performing the Lead-to-Cash Process — Focus on Sales

Lesson	Title
Topic 1	Understanding the Basics of the Sales Process
Topic 2	Using Master Data in Sales
Topic 3	Describing Presales Activities
Topic 4	Describing Sales Order Management
Topic 5	Describing the Delivery and Shipping Process
Topic 6	Describing Customer Billing
Topic 7	Managing Customer Payments

Unit: Performing the Lead-to-Cash Process — Focus on Service

Lesson	Title
Topic 8	Understanding the Basics of the Service Process
Topic 9	Exploring Master Data in Service
Topic 10	Describing Service Contract and Invoice Management
Topic 11	Managing Recurring Service
Topic 12	Service Analytics

Course reference: SAP Learning, "Exploring End-to-End Business Processes in SAP Business Suite," <https://learning.sap.com/courses/exploring-end-to-end-business-processes-in-sap-business-suite>

3. The Case: The Fresh Valley Refrigeration Deal

Iqhwa Cold Chain Systems (Pty) Ltd designs and manufactures commercial refrigeration units, sold through direct sales teams, an online configurator, and regional dealer partners, operating under a single company code with a Mkhambathini sales organisation and a Malagasy sales organisation, each selling across the "Refrigeration Systems" and "Parts & Accessories" divisions. Iqhwa Cold Chain Systems (Pty) Ltd has just closed the largest deal in its history: a contract with Fresh Valley Supermarkets, a supermarket chain, to supply 75 Iqhwa WI-500 units — and, critically, to keep that installation running for the next 3 years.

This is what makes the Fresh Valley Supermarkets deal a genuine Lead-to-Cash story rather than a simple sale. Fresh Valley Supermarkets is not a one-time buyer. Its 75 refrigeration units will run daily across stores across the Fresh Valley chain, and every one of them will eventually need a temperature-log and compressor check, or an unplanned repair after a store operations manager reports a fault. Iqhwa Cold Chain Systems (Pty) Ltd's revenue from Fresh Valley Supermarkets therefore comes in two connected streams: the original sale of the refrigeration units (Sales), and the ongoing contract to service and maintain them (Service). Both streams need to be visible to the same account team, use the same customer and product records, and ultimately post to the same set of accounts — which is exactly the integration problem SAP's Lead-to-Cash processes are designed to solve.

Appendix A records the ten business events of the Sales side of the deal — from first contact to final payment. Appendix B records the eleven business events of the Service side — from the moment the refrigeration units go live to the quarterly review of how well Iqhwa Cold Chain Systems (Pty) Ltd is servicing them. Read both logs carefully before starting Section 5: several tasks ask you to work across both.

A note on the scenario

Iqhwa Cold Chain Systems (Pty) Ltd and all names, figures and events in this case are fictional and created for this assignment. Iqhwa Cold Chain Systems (Pty) Ltd is designed as an extension of the "bike company" scenario used within the SAP Learning course itself, so that concepts you met in the course — sales areas, material types, business partners, equipment records, and so on — carry directly into this case.

Appendix A — Sales Event Log: Fresh Valley Supermarkets, the Order

#	Business Event	Relevant Lesson
1	A procurement manager at Fresh Valley Supermarkets contacts Iqhwa Cold Chain Systems (Pty) Ltd about equipping new stores with commercial refrigeration units. An Iqhwa Cold Chain Systems (Pty) Ltd sales representative logs the lead and opens an opportunity in SAP Sales Cloud.	Presales (Topic 3)
2	The sales rep prepares and sends Fresh Valley Supermarkets a formal sales quotation for 75 Iqhwa WI-500 units, delivered in 3 batches of 25, alongside a proposed 3-year service and maintenance contract.	Presales (Topic 3)
3	Fresh Valley Supermarkets is set up as a new customer: an existing FI Customer business partner is extended with the Customer (Sales) role for the wholesale sales area (Mkhambathini Sales • Wholesale • Refrigeration Systems).	Master Data (Topic 2)

#	Business Event	Relevant Lesson
4	Fresh Valley Supermarkets accepts the quotation and places a sales order for 75 Iqhwa WI-500 units, referencing the agreed 3-batch delivery schedule.	Order Management (Topic 4)
5	The system checks availability against Eston stock, confirms delivery dates for each batch, and applies the wholesale sales area's shipping conditions, which allow partial deliveries.	Order Management (Topic 4)
6	The warehouse creates the outbound delivery for the first batch of 25 refrigeration units, picks and packs them, and posts goods issue.	Delivery & Shipping (Topic 5)
7	A billing document is created for the first batch. An invoice is generated as PDF output and sent to Fresh Valley Supermarkets's accounts department.	Customer Billing (Topic 6)
8	Fresh Valley Supermarkets's payment for the first batch arrives by bank transfer thirty days later and is matched against the open invoice, clearing the receivable.	Payments (Topic 7)
9	The same delivery–billing–payment cycle (Events 6–8) repeats for the remaining batches of 25 refrigeration units over the following 2 months, completing the 75-unit order.	Delivery, Billing & Payments (Topic 5–7)
10	At quarter-end, Iqhwa Cold Chain Systems (Pty) Ltd's finance team reconciles the full Fresh Valley Supermarkets sales revenue from all 3 batches as part of month-end close.	Basics of the Sales Process (Topic 1)

Appendix B — Service Event Log: Fresh Valley Supermarkets, the Installation

#	Business Event	Relevant Lesson
11	As the refrigeration units are delivered, Iqhwa Cold Chain Systems (Pty) Ltd's service team creates an equipment record for each of the 75 refrigeration units, linking each serial number to Fresh Valley Supermarkets as the installed-base customer.	Master Data in Service (Topic 9)
12	A 3-year service contract is created for Fresh Valley Supermarkets, defining the covered maintenance scope, response-time commitments, and a monthly maintenance plan for the installation.	Service Contract & Invoice Management (Topic 10)
13	The service contract's billing plan generates a recurring monthly service invoice for Fresh Valley Supermarkets, separate from the original refrigeration unit sale invoices raised in Appendix A.	Service Contract & Invoice Management (Topic 10)
14	Following the maintenance plan, Iqhwa Cold Chain Systems (Pty) Ltd's scheduling tool automatically creates a batch of service orders for a routine monthly temperature-log and compressor check across the Fresh Valley Supermarkets installation.	Managing Recurring Service (Topic 11)
15	Field service technicians complete the routine checks, recording time confirmations against each service order and any parts consumed, such as door gasket seals.	Managing Recurring Service (Topic 11)
16	Independently, a store operations manager reports a fault with one refrigeration unit's compressor unit through the Fresh Valley Supermarkets operations app. A service notification is created in SAP describing the malfunction.	Basics of the Service Process (Topic 8)
17	A service technician reviews the notification and creates a service order, planning the required spare part — a replacement compressor unit — and the labour needed.	Basics of the Service Process (Topic 8)
18	The service order is checked and released: stock of the replacement part is confirmed sufficient and no further safety clearance is required before work can begin.	Basics of the Service Process (Topic 8)
19	The technician completes the repair on site, confirms the work performed and the part withdrawn from inventory, and the service order is technically completed.	Basics of the Service Process (Topic 8)
20	The completed service order is settled. Because the repair falls within the installation's service contract, it is billed against the contract rather than invoiced separately to Fresh Valley Supermarkets.	Service Contract & Invoice Management (Topic 10)
21	At quarter-end, Iqhwa Cold Chain Systems (Pty) Ltd's service manager reviews service analytics — average repair time, first-time-fix rate, and the profitability of the Fresh Valley Supermarkets contract — to report to leadership.	Service Analytics (Topic 12)

5. Your Tasks — What to Submit

Complete Parts A to F below, in order. Parts A and B use only Appendix A (Sales). Parts C and D use only Appendix B (Service). Part E uses both. Each part opens with a "Submit:" list of numbered sub-questions (e.g., A1, A2) — the mark shown after each one is that sub-question's share of the part's total, and the two always add up.

Part A — Mapping Sales Events to the Lead-to-Cash (Sales) Process (20 marks)

Submit:

A1 (14 marks) One table, 10 rows — one per event in Appendix A — with these columns: Event #, Sales Process Stage (Master Data, Presales, Order Processing, Product Delivery, Invoicing, or Payment Processing), Key SAP Document/Object, Responsible Role/System.

A2 (6 marks) For 3 of those 10 events (your choice), a 2–3 sentence justification explaining your classification, using SAP terms from the course.

Use the worked example below as your format guide for A1 — Event 1 is done for you.

#	Sales Stage	Key SAP Document/Object	Responsible Role/System	Justification
1	Presales	Lead / Opportunity	Sales rep, SAP Sales Cloud	The enquiry has not yet become a firm commitment; SAP Sales Cloud is the CRM front end used for presales in the course.

Marking focus: accuracy of classification across all 10 events (A1); correct and specific use of SAP terminology; quality of the three written justifications (A2).

Part B — Sales Process Flow, Master Data & Enterprise Structure (20 marks)

Submit:

B1 (10 marks) One swimlane diagram of the full sales cycle (Events 1–10, lead to final payment). One lane per department, role, or system. Show every step, the SAP document it creates, and the hand-off from SAP Sales Cloud (presales) to SAP S/4HANA Sales and Distribution.

B2 (5 marks) One table of the master data the process needs — material master, customer master (state which business partner role(s)), condition records — each with a 1-sentence reason.

B3 (5 marks) One table of the enterprise structure elements involved — company code, sales organisation, distribution channel, division, sales area, plant, storage location, shipping point — with example values from the case.

Marking focus: completeness and logical sequencing of the diagram (B1); correct SAP documents at each step; clear hand-offs; completeness and correctness of the master data (B2) and enterprise structure (B3) tables.

Part C — Mapping Service Events to the Lead-to-Cash (Service) Process (20 marks)

Submit:

C1 (14 marks) One table, 11 rows — one per event in Appendix B — with these columns: Event #, Service Process Stage (Master Data in Service, Service Contract & Invoice Management, Managing Recurring Service, Service Analytics, or a named

corrective-maintenance stage — Notification, Order Creation, Release, Execution, Confirmation, or Settlement), Key SAP Document/Object, Responsible Role/System, and Planned or Corrective.

C2 (6 marks) For 3 of those 11 events (your choice), a 2–3 sentence justification explaining your classification.

Marking focus: accuracy of classification across all 11 events (C1); correct distinction between planned and corrective maintenance; correct and specific use of SAP terminology; quality of the three written justifications (C2).

Part D — Service Process Flow, Master Data & Organizational Structure (20 marks)

Submit:

D1 (7 marks) Diagram 1: the corrective maintenance flow for the reported fault (Events 16–19) — notification through to technical completion.

D2 (7 marks) Diagram 2: the recurring maintenance flow for the routine check (Events 14–15) — maintenance plan through to confirmation.

D3 (4 marks) One table of the organizational/master data elements involved — Business Partner (customer and equipment), Maintenance Work Center, Personnel Number, SAP User, Service Contract — each with a 1-sentence reason.

D4 (2 marks) One short paragraph (100–150 words) comparing these elements to the sales-side enterprise structure you identified in Part B.

Marking focus: completeness and correctness of both diagrams (D1, D2); correct organizational/master data elements identified (D3); quality of the comparison with Part B (D4).

Part E — Sales-to-Service Integration and Critical Analysis (15 marks)

Submit:

E1 (5 marks) A short explanation, citing specific event numbers, of how the sales delivery (Events 6 and 9) leads to the service equipment records (Event 11), and how the recurring service contract billing (Event 13) differs from the original sales billing (Event 7).

E2 (6 marks) Two specific "what goes wrong" scenarios if the company ran Sales and Service on disconnected systems (for example, a technician unable to confirm warranty cover, or a customer billed twice for one repair), and how SAP's integrated Sales Cloud/Service Cloud/S/4HANA landscape prevents each one.

E3 (4 marks) A reflection, maximum 300 words, on why after-sales service — not just the initial sale — matters to the Lead-to-Cash revenue story for a company like this one.

Marking focus: depth and accuracy of the integration explanation (E1); specificity and correctness of the two integration-failure scenarios (E2); critical depth of the reflection (E3); word limit respected.

Part F — Professionalism and Presentation (5 marks)

Submit:

F1 (5 marks) Nothing new to write — this item marks the quality of the report you have already produced for Parts A–E: a clear structure with headings, correct and consistent SAP terminology, legible and properly labelled diagrams, consistent referencing (Harvard or APA) of the SAP Learning course, and a completed, signed Appendix D.

Marking focus: logical structure and clarity; correct and consistent SAP terminology; diagram legibility and labelling; consistent referencing; a complete contribution statement.

6. Deliverables and Format

- One written report per group, addressing Parts A–F in Section 5, in that order.
- Recommended length: 3,000–4,000 words, excluding diagrams, tables and appendices.
- Submit as a single PDF file to makombef@dut.ac.za, sent by the Group Leader, by 5 October 2026.
- Formatting: 11pt Arial, 1.15 line spacing, numbered pages, and a cover page stating your group number and every member's name and student ID.
- All diagrams must be your group's own original work — do not copy graphics from the SAP Learning course.
- Include Appendix D: a signed individual contribution statement listing each member's name, student ID, a brief summary of their contribution, and an approximate percentage contribution.
- Group size is 4–5 students. If your cohort does not divide evenly, speak to your instructor about an exception before you begin.

7. Marking Rubric

Marks are allocated as follows. Within each part, work is assessed on a graduated scale; the descriptor represents the standard required for full marks in that part.

Assessment Criterion	Marks	What Full Marks Looks Like
Part A — Sales Process Mapping	20	All 10 sales events correctly classified by stage, document/object and responsible role; three justifications are specific and use SAP terminology accurately.
Part B — Sales Flow, Master Data & Structure	20	A complete, logically sequenced sales swimlane diagram with correct documents and clear hand-offs; complete and correct master data and enterprise structure tables.
Part C — Service Process Mapping	20	All 11 service events correctly classified by stage, document/object and role, with planned vs. corrective maintenance correctly distinguished; three justifications are specific and accurate.
Part D — Service Flow, Master Data & Structure	20	Both the corrective and recurring maintenance diagrams are complete and correctly sequenced; organizational/master data elements are correct; the comparison with Part B is insightful.
Part E — Sales-to-Service Integration	15	The Sales–Service connection is explained accurately with specific event references; both integration-failure scenarios are plausible and well argued; reflection is critical and within the word limit.
Part F — Professionalism	5	Report is clearly structured, consistently and correctly uses SAP terminology, diagrams are legible and properly labelled, referencing is consistent, and the contribution statement is complete.
Total	100	

8. Academic Integrity

All work submitted must be your group's own. You may discuss the case with other groups in general terms, but your event classifications, diagrams, tables and written analysis must be produced independently by your group. Cite the SAP Learning course and any other source you draw on, using a consistent referencing style (Harvard or APA). The use of generative AI tools should be disclosed, particularly how and where they were used. Suspected plagiarism, collusion between groups, or fabricated contribution statements will be referred under the Durban University of Technology academic integrity procedures.

Appendix C — Glossary of Key SAP Terms

This glossary is provided for reference. It is not exhaustive — you are expected to draw on the full depth of terminology introduced in both units.

Term	Definition
Lead / Opportunity	An early-stage record of customer interest, managed in SAP Sales Cloud, before a firm order exists.
Customer Master (FI Customer role)	The accounting view of a customer business partner, valid for a company code; holds the reconciliation account used in the general ledger.
Customer Master (Customer / Sales role)	The sales view of a customer business partner, valid for a sales area; holds shipping, billing and pricing information.
Sales Area	The combination of sales organisation, distribution channel and division within which a sales document is processed.
Condition Record	A stored pricing element (price, discount, freight, or tax) used by a pricing procedure to price a document.
Sales Order / Delivery / Billing Document	The documents that record, respectively, a confirmed customer order, the physical shipment of goods, and the invoice raised for it.
Equipment / Installed Base	A master record representing a specific physical unit tracked individually for service and maintenance once it has been sold.
Functional Location	A structured record of where a piece of equipment operates, used to organise maintenance by location or asset grouping.
Business Partner (Service)	The unified master data object, shared across sales and service, that represents a customer, contact, or piece of equipment in a service context.
Service Contract	An agreement defining the scope, response times and billing terms for ongoing service or maintenance of a customer's products.
Maintenance Plan	A scheduling object that automatically generates service orders for planned, recurring maintenance activity.
Service Notification	The document created to record a reported malfunction or service request, triggering the corrective maintenance process.
Service Order	The document that authorises and tracks the labour, parts and cost of a specific service or repair activity.
Confirmation	The record of work actually performed against a service order, including time spent and parts consumed.
Settlement	The step that posts the completed cost of a service order to its correct destination — a contract, a cost centre, or a customer invoice.
Maintenance Work Center	An organizational unit representing a group of technicians or a capacity resource that performs service work.
Personnel Number	The unique identifier for an individual employee, such as a field service technician, within SAP.
Enterprise Asset Management (EAM)	The broader SAP framework for planning, executing and analysing maintenance activity across physical assets.

Term	Definition
Service Analytics	Reporting on service performance — for example, repair time, first-time-fix rate, and contract profitability.

Appendix D — Individual Contribution Statement

Every group member must sign this statement before submission. Reproduce the table below in your report, completed for every member of your group.

Name	Student ID	Summary of Contribution	Approx. %	Signature